

Hill Climbing

A Basic Optimization Algorithm

At a Glance

Applicability: Discrete Optimization Problems

Abstract Description: Start at one point, go towards the best neighbour, repeat.

Parameters:

- Starting point: Which point to start on?
- Neighbours: Which points are considered neighbours?

The Algorithm

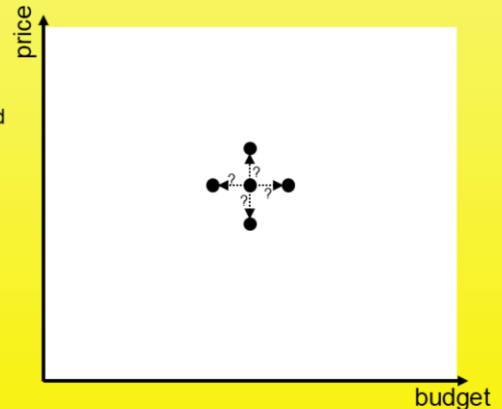
1) Start at a point and find all neighbours

- Traditionally +1 and -1 for each dimension
- Other neighbourhoods also possible

2) Find best neighbour

- If no neighbour is better than the current point, we are finished

3) Move to best neighbour and repeat from 1



The algorithm walks around in the state space, always trying to walk to a better neighbour. It is initialized with a starting point. From there, it looks at all neighbours and chooses the best one to walk to. This is repeated until there is no way to improve. Visualized in a search space, this means the algorithm always moves uphill until it is at a peak. This is where the name “Hill Climbing” comes from.

In more detail, this is algorithm works as follows:

- 1) Start from the current point and search for all neighbors. The easiest way to determine neighbors is to look at the previous and next value of each dimension. E.g., in a two-dimensional space, like the one visualized on the slide, the neighbors of the point (500,300) are (499,300) and (501,300) for the first parameter and (500,299) and (500,301) for the second parameters. These are the four points in the visualization. It is possible to define the neighborhood differently, e.g., by combining changes of the parameters (e.g., including the point (501,301) into the neighborhood) or by making bigger steps.
- 2) Finding the best neighbor is a matter of iterating over all neighbors and remembering the best one. If no neighbor is better than the current point, then the algorithm is finished.
- 3) If we're not finished, we repeat from step 1 with the best neighbor as new starting point.